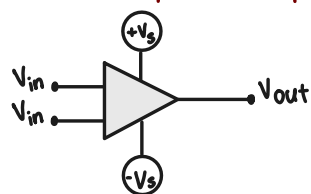


Operational Amplifiers (op-amps)

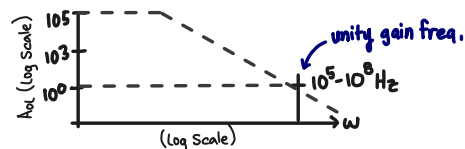


$$V_{out} = A_{OL}(V_{in+} - V_{in-})$$

$$A_{OL} \approx 10^4 - 10^6 \gg 1$$

Properties of op-amps

- Inputs are very high impedance ($10^6 - 10^{14} \Omega$) (typically assume $Z_{in} \approx \infty$ [1-100 Ω])
- Output impedance is low (1-100 Ω) (assume Z_{out} is negligible)
- output $-V_s < V_{out} < +V_s$ (typically assume A_{OL} is large $\rightarrow \infty$)



In a non-Inverting Amplifier

$$V_{in+} = V_{in}, V_{in-} = \frac{R_2}{R_1 + R_2} \cdot V_{out} = \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{R_1 + R_2}{R_2} \right) \cdot V_{in}$$

$$V_{in-} \approx V_{in}$$

This is generally true for op-amp circuits w/ negative feedback

Golden Rules for op-amp circuits w/ Negative Feedback

- If there is a resistor (or other circuit element) connecting output back to inverting input:

- The output will do whatever is necessary to make inputs equal

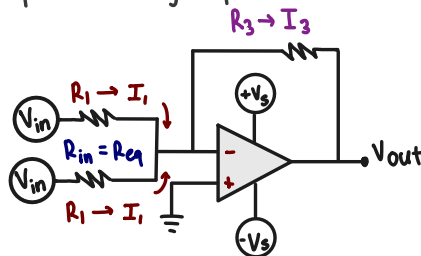
$$* V_{in+} = V_{in-}$$

↳ If non-Inverting input is grounded then the inverting input will be called a 'virtual ground'

- Also assume no current flows into inputs $\Rightarrow$ assume Z_{in} is low

Solve V_{out} for both examples

example 1: Summing Amplifier



$$V_{in+} = V_{in-} = 0$$

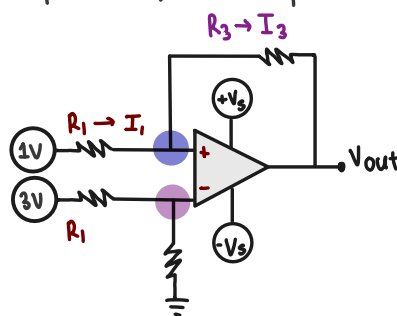
↳ $I_1 + I_2 = I_3$ Junction Rule

$$I_1 = \frac{V_{in1}}{R_1}, I_2 = \frac{V_{in2}}{R_2}, I_3 = \frac{V_{out}}{R_3}$$

$$\text{so: } V_{out} = -(I_1 + I_2)R_3 = -\left[\frac{(V_{in1} + V_{in2})}{R_1} \right] \cdot R_3$$

$$\Rightarrow V_{out} = -\left(\frac{R_3}{R_1} \right) (V_{in1} + V_{in2})$$

example 1: Differential Amplifier



$$R_1 = 500 \Omega, R_2 = 1000 \Omega$$

$$V_{in+} = V_{in-}$$

$$V_{in} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right) = 3 \left(\frac{1000}{1500} \right) = 2V$$

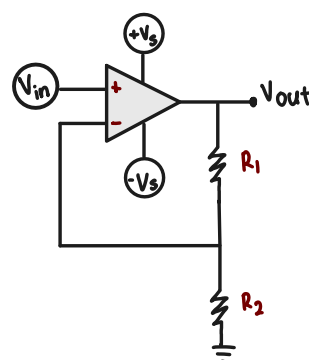
$$I_1 = \frac{1V - 2V}{500 \Omega} = -2mA \Rightarrow I_1 R_2 = 2 - V_{out}$$

$$\Rightarrow -2(1000) = 2 - V_{out}$$

$$-2 = 2 - V_{out} \Rightarrow V_{out} = 4V$$

$$V_{out} = \frac{-R_2}{R_1} (V_{in1} - V_{in2})$$

Op-amp circuit w/ negative feedback



$$V_{out} = A_{OL}(V_{in+} - V_{out-})$$

assume V_{in} draws zero current

$$\text{then } V_{in} = \frac{R_2}{R_1 + R_2} \cdot V_{out} \quad \& \quad V_{out} = A_{OL} \left(V_{in} - \frac{R_2}{R_1 + R_2} \right)$$

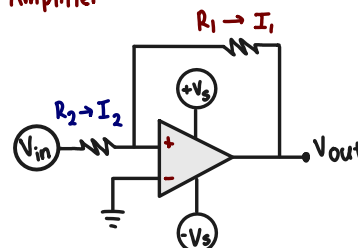
$$\Rightarrow V_{out} \left(1 + A_{OL} \frac{R_2}{R_1 + R_2} \right) = A_{OL} \cdot V_{in} \Rightarrow V_{out} = \left(\frac{A_{OL}}{1 + A_{OL} \left(\frac{R_2}{R_1 + R_2} \right)} \right) V_{in}$$

Gain of Amplifier - set by $R_1 + R_2$
↳ not by A_{OL}

Can V_{in} not be railed? - yes!

$$\text{If } A_{OL} \gg 1: V_{out} \approx \left(\frac{A_{OL}}{A_{OL} \frac{R_2}{R_1 + R_2}} \right) V_{in} = \left(\frac{1}{\frac{R_2}{R_1 + R_2}} \right) V_{in} = \left(1 + \frac{R_1}{R_2} \right) V_{in}$$

Inverting Amplifier



• If $V_{in} > 0$, what is V_{out} ?

If V_{in} increases a little bit
 $\rightarrow V_{out}$ wants to decrease a lot

$\Rightarrow$ but now current flows through R_1 & R_2
↳ so, $I_1 = I_2$ (no current drawn by inputs)

this current decreases V_{in} by $I_2 \cdot R_2$
• If V_{out} kept decreasing eventually $V_{in} \rightarrow 0$

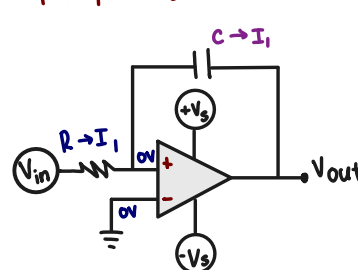
Circuit reaches equilibrium @ $V_{in-} \approx V_{in+}$

$$V_{in} = V_{in+} = V_{in-} = V_{out}$$

$$V_{out} = V_{in} \text{ unity Gain Buffer}$$

Get better I/O impedance characteristics
(even if finite R_{out}): $V_{out} = V_{in}$

Op-Amp Integrator



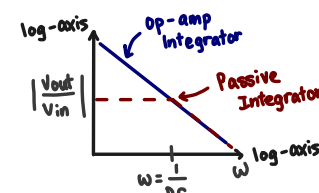
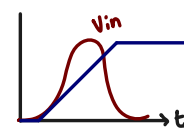
$$V_{in} = 0, V_{out} = -V_c = \frac{-Q}{C} = \frac{-1}{C} \int_{t=0}^t I_1 dt' = \frac{-1}{RC} \int_{t=0}^t V_{in}(t) dt$$

- Looks like an integrator

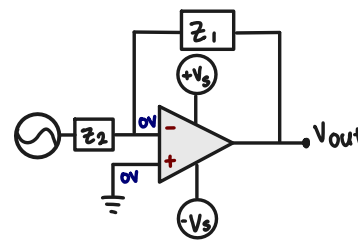
* watch out! A small DC Voltage
@ input will eventually integrate upto
large output voltage

Compare to passive Integrator

Looks like integrator
only when $\omega \gg \frac{1}{RC}$



Frequency Dependent Amplifier



$$\text{In General: } V_{out} = \frac{-Z_1}{Z_2} \cdot V_{in}$$

$$\text{ex: Integrator - } Z_1 = \frac{1}{i\omega C}, Z_2 = R$$

$$V_{out} = \frac{-1}{i\omega RC} \cdot V_{in}$$

